

Analyzing Ellipses: Standard Form and Graphing

Precalculus / Algebra 2 Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Find the center, foci, and vertices of an ellipse given its foci and major axis length
- Derive the standard equation of an ellipse using the values of a , b , and the center
- Graph an ellipse by identifying its key features including center, vertices, and foci

Problems

1. An ellipse has foci at $(0, 1)$ and $(4, 1)$. Find the center of the ellipse.

$$F_1 = (0, 1), \quad F_2 = (4, 1)$$

2. An ellipse has foci at $(0, 1)$ and $(4, 1)$. Find the value of c , the distance from the center to each focus.

$$F_1 = (0, 1), \quad F_2 = (4, 1), \quad \text{Center} = (2, 1)$$

3. An ellipse has a major axis of length 6. Find the value of a .

$$2a = 6$$

4. Using the values c equals 2 and a equals 3, find the value of b squared.

$$c^2 = a^2 - b^2$$

5. Write the standard equation of the ellipse with center $(2, 1)$, a squared equals 9, and b squared equals 5, where the major axis is horizontal.

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$$

6. Find the coordinates of the two horizontal vertices of the ellipse below.



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$$\frac{(x-2)^2}{9} + \frac{(y-1)^2}{5} = 1$$

7. Find the coordinates of the two vertical co-vertices of the ellipse below.

$$\frac{(x-2)^2}{9} + \frac{(y-1)^2}{5} = 1$$

8. An ellipse has foci at (2, 0) and (2, 4) and a major axis of length 10. Find the center, the values of a, c, and b squared.

$$F_1 = (2, 0), \quad F_2 = (2, 4), \quad \text{major axis} = 10$$

9. Write the standard equation of the ellipse with foci at (2, 0) and (2, 4) and major axis of length 10. The major axis is vertical.

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$

10. Given the standard equation of an ellipse below, identify the center, foci, and vertices, then describe the orientation of the major axis.

$$\frac{(x+3)^2}{16} + \frac{(y-5)^2}{25} = 1$$



Analyzing Ellipses: Standard Form and Graphing — Answer Key

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Answer Key

1. Answer: (2, 1)

- The center is the midpoint of the two foci.
- Center = $\left(\frac{0+4}{2}, \frac{1+1}{2}\right) = (2, 1)$

2. Answer: $c = 2$

- c is the distance from the center (2, 1) to either focus.
- $c = |2 - 0| = 2$

3. Answer: $a = 3$

- The major axis length equals $2a$.
- $2a = 6 \rightarrow a = 3$

4. Answer: $b^2 = 5$

- Substitute $c = 2$ and $a = 3$ into the formula.
- $4 = 9 - b^2 \rightarrow b^2 = 9 - 4 = 5$

5. Answer: $(x-2)^2 / 9 + (y-1)^2 / 5 = 1$

- Substitute $h = 2$, $k = 1$, $a^2 = 9$, $b^2 = 5$.
- $\frac{(x-2)^2}{9} + \frac{(y-1)^2}{5} = 1$

6. Answer: (-1, 1) and (5, 1)

- The horizontal vertices are found by moving $a = 3$ units left and right from the center (2, 1).
- Left vertex: $(2 - 3, 1) = (-1, 1)$; Right vertex: $(2 + 3, 1) = (5, 1)$

7. Answer: (2, 1 + $\sqrt{5}$) and (2, 1 - $\sqrt{5}$)

- The vertical co-vertices are found by moving $b = \sqrt{5} \approx 2.24$ units up and down from center (2, 1).
- Top: $(2, 1 + \sqrt{5})$; Bottom: $(2, 1 - \sqrt{5})$

8. Answer: Center = (2, 2); $a = 5$; $c = 2$; $b^2 = 21$

- Center = midpoint of foci = $\left(\frac{2+2}{2}, \frac{0+4}{2}\right) = (2, 2)$
- $2a = 10 \rightarrow a = 5$; $c = |2 - 0| = 2$
- $b^2 = a^2 - c^2 = 25 - 4 = 21$

9. Answer: $(x-2)^2 / 21 + (y-2)^2 / 25 = 1$

- Center = (2, 2), $a^2 = 25$, $b^2 = 21$. Since the major axis is vertical, a^2 goes under the y term.
- $\frac{(x-2)^2}{21} + \frac{(y-2)^2}{25} = 1$



- Center: $h = -3$, $k = 5$. Since $25 > 16$, the major axis is vertical, so $a^2 = 25$ and $b^2 = 16$.
 - $a = 5$, $b = 4$. Vertices: $(-3, 5+5) = (-3, 10)$ and $(-3, 5-5) = (-3, 0)$.
 - $c^2 = a^2 - b^2 = 25 - 16 = 9 \rightarrow c = 3$. Foci: $(-3, 5+3) = (-3, 8)$ and $(-3, 5-3) = (-3, 2)$.
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